

Assignment 5

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1 Features vs Jump

1.1 Raw Variables

Raw Variable	Description
jump	Target variable representing sudden large changes in asset price or returns.
industry	Categorical variable for asset industry, numerically encoded.
beta	Firm's beta, sensitivity to market movements.
return_vol_6months	Historical volatility of firm's returns over past 6 months.
universe_vol_6months	Market/benchmark volatility over past 6 months.
FEDFUNDS	Effective federal funds rate.
M2REAL	Inflation-adjusted M2 money supply.
T10Y2YM	10-year minus 2-year Treasury yield spread.
UNRATE	Unemployment rate.
USREC	US recession indicator (binary).
VIXCLS_last	Latest VIX index value (end of month).
credit_risk	Measure of market credit risk (BAA-AAA spread).

Table 1: Explanation of Raw Variables

The data is aggregated month-wise, and the percentage of jumps (return > 0.1) is calculated and plotted against macro variables to assess correlations.

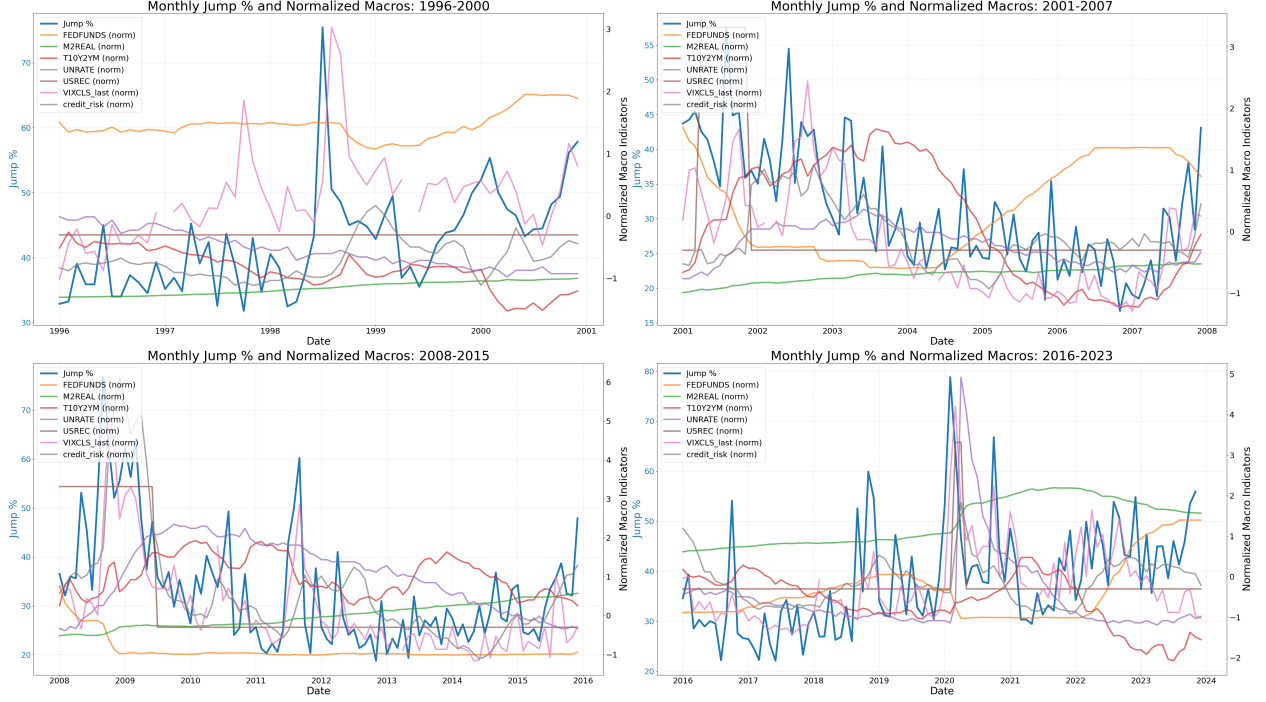


Figure 1: Monthly Jump Percentage vs Macros. VIXCLS shows the strongest correlation with jumps.

Lagging macro variables by one month allows analysis of how previous-month conditions affect current jump rates. Lagged VIXCLS shows the highest correlation (0.5) with jumps, while FEDFUNDS, USREC, T10Y2YM, and credit risk show moderate associations.

1.2 Feature Engineering

By using z-score standardized and lagged features (Table 2), the model can detect and quantify unusually large changes in asset behavior. Z-scores highlight extreme deviations from historical norms, such as spikes in returns, prices, or trading volume, signaling potential jumps. Lagged features provide temporal context, capturing patterns in recent periods that often precede jumps. Together, they allow the model to focus on abnormal movements relative to recent history, improving prediction of rare but significant jumps.

Variable	Description
VIXCLS_last	Most recent VIX index value.
beta_prev	Previous period asset beta.
return_vol_6months_prev	Historical return volatility from previous period.
universe_vol_6months_prev	Market volatility from previous period.
PRC_prev	Previous period closing price.
VOL_prev	Previous period trading volume.
SHROUT_prev	Previous period shares outstanding.
RETX_prev	Previous period return.
vwretd_prev	Previous period value-weighted market return.
UNRATE_prev	Previous period unemployment rate.
FEDFUNDS_prev	Previous period federal funds rate.
M2REAL_prev	Previous period real M2.
credit_risk_prev	Previous period credit risk.
USREC_prev	Previous period US recession indicator.
SICCD_encoded	Numerically encoded industry classification.
RETX_prev_z_3m	Z-score of past 3-month return.
RETX_prev_z_6m	Z-score of past 6-month return.
RETX_prev_z_12m	Z-score of past 12-month return.
PRC_prev_z_3m	Z-score of past 3-month price.
PRC_prev_z_6m	Z-score of past 6-month price.
PRC_prev_z_12m	Z-score of past 12-month price.
VOL_prev_z_3m	Z-score of past 3-month volume.
VOL_prev_z_6m	Z-score of past 6-month volume.
VOL_prev_z_12m	Z-score of past 12-month volume.
excess_return_prev	Previous period excess return.
beta_VIX_interaction	Interaction term of beta and VIX.
dollar_trading_vol_prev	Previous period dollar trading volume.

Table 2: Previous-Period and Derived Variables

1.3 Correlation Plots

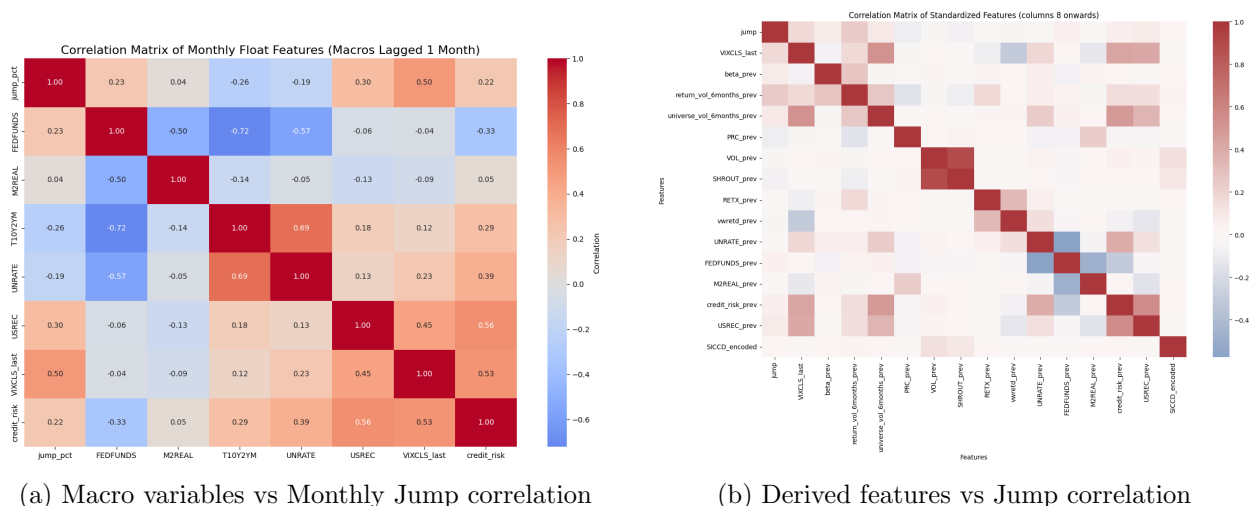


Figure 2: Correlation plots of macro variables and engineered features with monthly jumps.

VIXCLS_Last, return_vol_6months_prev, beta, credit_risk, USREC_prev are good predictors. Although some of these are highly correlated between themselves

2 Models and Performance

2.1 Data Preparation

Three datasets were prepared:

- **Fixed split:** Training on all data up to 2017, testing from 2018 onward.
- **Rolling out-of-sample:** Train on all data before a target year, test on that year, evaluating year-by-year predictive performance.
- **Moving window:** Train on a fixed-length (e.g., 22-year) window preceding the test year, testing on the next year.

All features were standardized using z-score normalization.

2.2 Logistic Regression

Table 3: Logistic Regression Performance (KS, AUC, Accuracy)

Split / Year	KS Train	KS Test	AUC Train	AUC Test	Train Acc	Test Acc
Fixed split	0.2967	0.2819	0.7051	0.6798	0.7227	0.6955
Rolling year 2018	0.2967	0.2745	0.7051	0.6878	0.7227	0.7404
Rolling year 2019	0.2947	0.2553	0.7046	0.6704	0.7232	0.7534
Rolling year 2020	0.2918	0.2936	0.7033	0.6795	0.7248	0.6403
Rolling year 2021	0.2961	0.2821	0.7038	0.6660	0.7213	0.7432
Rolling year 2022	0.2969	0.2552	0.7018	0.6462	0.7217	0.6534
Rolling year 2023	0.2950	0.2684	0.6995	0.6692	0.7188	0.6570

2.3 Post-LASSO, RidgeCV, KNN, and XGBoost

I chose the fixed sample (1996-2017 end) for training and testing on data from 2018-2023 for the following models 1. L1 Logistic Regression 2. L2 Logistic Regression 3. KNN 4. XgBoost

for KNN and XgBoost I used the gridsearchcv and randomsearchcv classes available in the sklearn library to select the best hyperparameters which are mentioned in the table

Table 4: Performance Summary of Post-LASSO, RidgeCV, KNN, and XGBoost

Model	HyperParams	Train Acc	Test Acc	Train Misclass.	Test Misclass.	Train / Test AUC
Post-LASSO Logistic Regression	$\alpha = 0.005693$, features=[0,2,4,6,9,14,19,20,24,25]	0.7227	0.6968	0.2773	0.3032	0.7025 / 0.6837
RidgeCV	$\alpha = 483.293$, features=[0-26]	0.7204	0.6981	0.2796	0.3019	-
KNN	$n_neighbors = 49$	0.7298	0.6915	0.2702	0.3085	-
XGBoost	subsample=0.9, min_child_weight=9, max_depth=3, gamma=5, colsample_bytree=1.0	0.7492	0.6962	-	-	0.7644 / 0.6797